CarbonQuest: an India-calibrated climate twin for personal carbon footprints

Project write-up for EurekaDev 2026 | Coding Track | Category: Biology/Medical and Environmental Science

Author: [your name] | Live demo: [paste link] | Code: [paste GitHub link]

Abstract

Carbon calculators usually end with a single number and no clear next step. CarbonQuest is a web application that builds a "climate twin" of a person's year in India: the user describes commuting, flights, home electricity, diet, spending and waste, and the application returns an annual footprint in tonnes of CO_2e, a letter grade, a breakdown by source, a comparison with India, the world and a 1.5°C-aligned level, and a set of actions whose savings can be tested. A projection to 2050 shows the effect of acting now. All factors are documented and traceable to published sources, and limitations are stated inside the application. This document gives the problem, a short literature background, the solution outline, the method, example results and references.

1. Problem

Personal footprint tools tend to report a total and stop. Users learn that their footprint is, for example, 4 tonnes a year, but not which habit matters most, how they compare with a climate-compatible level, or what a specific change would save. Tools built around Western defaults also assume car-centred commutes and national grid factors that differ from India's. CarbonQuest addresses the question people actually ask: **what can I change, and does it matter?**

2. Background and literature review

Climate targets and benchmarks. The IPCC Sixth Assessment Report states that limiting warming to 1.5°C requires deep, rapid and sustained greenhouse-gas reductions [1]. Per-person emissions by country are compiled by the Global Carbon Project and presented by Our World in Data [2], which shows India well below the world average; CarbonQuest uses about 2.0 t for India and about 4.7 t for the world. The Hot or Cool Institute's *1.5-Degree Lifestyles* report translates carbon budgets into per-person lifestyle footprint levels and examines domains such as housing, mobility and food [3]; CarbonQuest uses about 2.3 t per person by 2030 as its 1.5°C-aligned reference.

Food. Poore and Nemecek's global meta-analysis found large differences in environmental impact between foods and producers, and showed that what people eat is a major lever [4]. Scarborough et al. compared dietary greenhouse-gas emissions of meat-eaters, fish-eaters, vegetarians and vegans in the UK and found clear differences between diet groups [5]. CarbonQuest therefore models diet as four annual levels (vegan, vegetarian, mixed, meat-heavy).

Electricity, transport and flights. Home electricity emissions depend on the grid. The Central Electricity Authority publishes the CO_2 Baseline Database for the Indian power sector, from which CarbonQuest takes about 0.71 kg CO_2 per kWh [6]. Transport and flight factors follow the UK government's GHG conversion factors (DEFRA) [7], adapted for Indian vehicles such as scooters and metro travel.

Gap addressed. The sources above provide factors and targets but not an interactive, India-specific way for an individual to see how habits compare with a climate-compatible level and what changes would save. CarbonQuest combines them in one transparent model.

3. Solution outline

CarbonQuest is a single, self-contained web page (HTML, CSS and JavaScript, no framework or server). Main parts:

Part	What it does
Lifestyle inputs	Commute distance and mode (petrol, diesel, electric car, motorbike or scooter, bus or metro, walk or cycle), short and long flights, home electricity, people sharing the home, diet, monthly spending on goods, and waste habits.
Personas	One tap loads Student, Metro professional, Frequent flyer or Eco champion so users can compare lifestyles instantly.
Footprint ring and grade	Annual footprint in t CO_2e, a grade from A to E, a plain-language message naming the biggest source, and the number of mature trees needed to absorb it.
Benchmarks and danger meter	Your footprint against India (about 2.0 t), world (about 4.7 t) and the 1.5°C level (about 2.3 t); the page colours and a danger label change with the result.
Where it comes from	Breakdown into travel, flights, home energy, food, and goods and waste.
Real actions and badges	Seven actions (for example bus or metro, electric vehicle, rooftop solar, vegetarian four days a week) show their yearly saving; badges reward progress.
Road to 2050	Compares doing nothing with applying the chosen actions by 2031, then a 3% yearly decline from grid cleaning.
Method and sources	Every factor and its source are listed in the application, with limitations.

4. Method

The annual footprint is the sum of five categories, in tonnes of CO_2e per person per year:

$$\mathbf{E = E_{travel} + E_{flights} + E_{home} + E_{food} + E_{goods\ and\ waste}}$$

Category	Formula used	Factors
Travel	weekly km x 52 x f_{mode} / 1000	Petrol car 0.15, diesel car 0.16, electric car 0.05, motorbike or scooter 0.04, bus or metro 0.03, walk or cycle 0 (kg CO_2e per km)
Flights	short flights x 0.2 + long flights x 1.6	0.2 t per short flight, 1.6 t per long flight
Home energy	kWh per month x 12 x 0.71 / 1000 / people sharing the home	0.71 kg CO_2 per kWh (Indian grid) [6]
Food	annual value by diet	Vegan 0.9, vegetarian 1.2, mixed 1.8, meat-heavy 2.8 t per year [4][5]
Goods and waste	monthly spend x 12 / 1000 x 0.008, plus waste term	0.008 t per Rs 1000 spent; waste 0.40 t (mostly landfill) or 0.15 t (recycle and compost)

Grades: A at or below 2.3 t, B up to 3.5 t, C up to 4.7 t, D up to 7 t, E above 7 t. Trees are estimated at about 21 kg CO_2 absorbed per mature tree per year. Each action reduces one category by a stated share, for example rooftop solar covers about 70% of home electricity, and buying less and second-hand cuts goods impact by 30%.

5. Example results

Running the four built-in personas through the model gives the footprints below. They illustrate how strongly flights, electricity and diet drive the total.

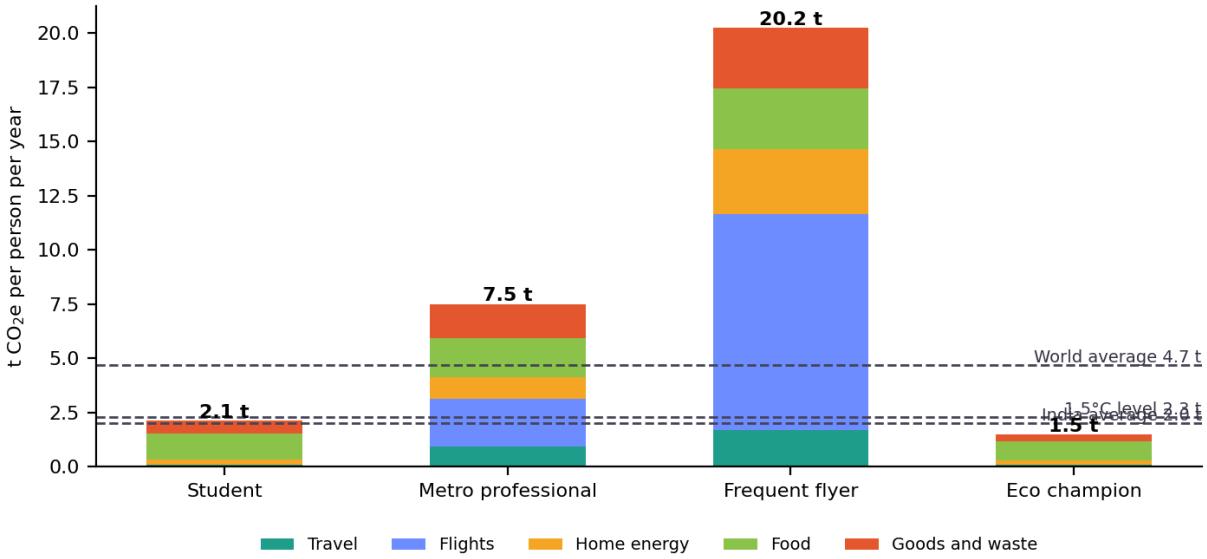


Figure 1. Annual footprint of the four personas by category, with benchmark levels. Values are produced by the same formulas as the application.

Persona	Travel	Flights	Home energy	Food	Goods and waste	Total (t)
Student	0.06	0.00	0.26	1.20	0.59	2.11
Metro professional	0.94	2.20	0.99	1.80	1.55	7.48
Frequent flyer	1.66	10.00	2.98	2.80	2.80	20.25
Eco champion	0.06	0.00	0.21	0.90	0.29	1.47

Reading the table: the Student and Eco champion fall below the 1.5°C-aligned level, while the Frequent flyer is several times above the world average, mainly because of flights and electricity. This is the kind of insight the action list is meant to turn into concrete steps.

6. Limitations

CarbonQuest is an educational estimator, not a certified audit. Factors are rounded averages for India and will differ from an individual's exact case. The model uses a single national grid factor, does not include embodied emissions in housing, and uses a rough average for spending. Diet values are annual per-person constants, and the 2050 pathway uses simple assumptions (actions applied by 2031, then 3% a year). Benchmark figures come from different years and mix CO_2 and CO_2e, so comparison lines are rough guides.

7. Future work

City- and state-level grid data, bill upload to replace sliders with real electricity and fuel numbers, Hindi and Marathi language support, saved progress with weekly challenges, and a mobile app.

References

[1] IPCC (2023). *Climate Change 2023: Synthesis Report. Contribution of Working Groups I, II and III to the Sixth Assessment Report.* Intergovernmental Panel on Climate Change, Geneva.

[2] Global Carbon Project, *Global Carbon Budget*; and Ritchie, H., Roser, M. and Rosado, P., *CO_2 and Greenhouse Gas Emissions*, Our World in Data (ourworldindata.org).

[3] Akenji, L. et al. (2021). *1.5-Degree Lifestyles: Towards A Fair Consumption Space for All.* Hot or Cool Institute, Berlin.

[4] Poore, J. and Nemecek, T. (2018). Reducing food's environmental impacts through producers and consumers. *Science*, 360(6392), 987–992.

[5] Scarborough, P. et al. (2014). Dietary greenhouse gas emissions of meat-eaters, fish-eaters, vegetarians and vegans in the UK. *Climatic Change*, 125, 179–192.

[6] Central Electricity Authority, Government of India. *CO_2 Baseline Database for the Indian Power Sector.*

[7] UK Government (DEFRA / DESNZ). *Greenhouse gas reporting: conversion factors.*